Md Mushfiqur Azam

210-895-8562 | mdmushfiqur.azam@my.utsa.edu | linkedin.com/in/mushfiqur-azam | github.com/Mushfiq5647

Summary

Ph.D. candidate in Computer Science (CGPA 4.0) specializing in egocentric vision and 3D human pose estimation. Author of 6 papers (4 first-author; CVPR-W, WACV-W, IEEE), with 2+ years prior experience engineering production-scale backend systems serving 4M+ users.

Education

The University of Texas at San Antonio — *San Antonio, TX, USA*
Ph.D. in Computer Science and Engineering — 2023 – Present
CGPA: ***4.00/4.00*** Advisor: Dr. Kevin Desai

Khulna University of Engineering & Technology — *Khulna, Bangladesh*
B.Sc. in Computer Science and Engineering — 2016 – 2020
CGPA: ***3.55/4.00*** (Last 60 credits: ***3.79/4.00***) Advisor: Dr. Sk. Md. Masudul Ahsan

Publications

1. **Md Mushfiqur Azam**, Kevin Desai. **"A Survey on 3D Egocentric Human Pose Estimation."** *IEEE/CVF CVPR Workshops*, 2024, pp. 1643–1654.
2. Ayda Eghbalian, Md Mushfiqur Azam, Katie Holloway, Leslie Neely, Kevin Desai. **"Applying Computer Vision to Analyze Self-Injurious Behaviors in Children with Autism Spectrum Disorder."** *IEEE/CVF WACV Workshops*, 2025, pp. 11–20.
3. Sadia Mubashshira, Md Mushfiqur Azam, Sk. Md. Masudul Ahsan. **"An Unsupervised Approach for Road Surface Crack Detection."** *IEEE Region 10 Symposium (TENSYMP)*, 2020.
4. **Md Mushfiqur Azam**, John Quarles, Kevin Desai. **"AG-EgoPose: Leveraging Action-Guided Motion and Kinematic Joint Encoding for Egocentric 3D Pose Estimation."** *(Under review).*
5. **Md Mushfiqur Azam**, Nipa Anjum, John Quarles, Kevin Desai. **"EgoBalanceFormer: Predicting Human Balance from Egocentric Pose and VR-Native Motion."** *IEEE ISMAR*, 2026, Bari, Italy (under review).
6. **Md Mushfiqur Azam**, John Quarles, Kevin Desai. **"TSR-Ego: Temporally Guided Stereo Refinement Framework for Egocentric 3D Human Pose Estimation."** *(Under review).*

Experience

Graduate Research Assistant — **Jan 2024 – Present**
The University of Texas at San Antonio — *San Antonio, TX*

- Research egocentric vision for 3D human understanding, including pose estimation, activity recognition, and human-scene interaction.
- Designed a transformer-based egocentric pose framework in PyTorch to address occlusion in real-world data, reducing error by **12%** and **9%** over prior baselines.
- Built **EgoBalanceFormer**, a multimodal transformer for VR balance prediction, cutting CoP RMSE by **11.0%** on CoP_X and **12.4%** on CoP_Y.
- Engineered the backend for a multi-user VR chemistry lab (Unity / C#) with LAN multiplayer and experiment logic.

Software Engineer — **Aug 2021 – Jul 2023**
Kona Software Lab Limited — *Dhaka, Bangladesh*

- Engineered core microservices (payments, notifications) in Spring Boot and Oracle DB for a leading mobile financial service supporting **4M+** users.
- Built a high-scale notification system delivering millions of alerts via APN, FCM, and HPK.
- Developed Angular web features (purchase history, biller services) over REST APIs, and a graph monitoring service surfacing hourly request-rate insights.

Junior Software Engineer — **Dec 2020 – Aug 2021**
Nascenia Limited — *Dhaka, Bangladesh*

- Designed and built the **Problem Solution Tree** feature (Ruby on Rails) automating an NGO's problem-solution workflow, and implemented platform-wide internationalization to broaden accessibility.

Technical Skills

Deep Learning / ML: PyTorch, TensorFlow, Transformers, Attention Mechanisms, Transfer Learning.
3D / Computer Vision: Egocentric Vision, 3D Pose Estimation, Human Mesh Recovery, SMPL / SMPL-X, OpenCV.
Data & Visualization: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter / Colab.
Engineering: Python, Java, Spring Boot, Angular, SQL, Oracle DB, Git.